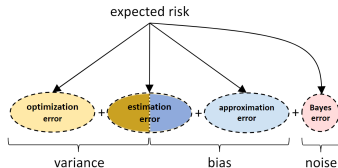


A 3x3 grid with a blue path starting at the top-left corner (0,0) and ending at the middle-right cell (1,2). The path consists of the following cells: (0,0), (0,1), (0,2), (1,2). Obstacles (X) are located at (0,2), (1,0), (2,0), and (2,1). Empty cells (O) are located at (0,1), (1,1), and (2,2).

Bias-Variance 2:

Approximation and Estimation error



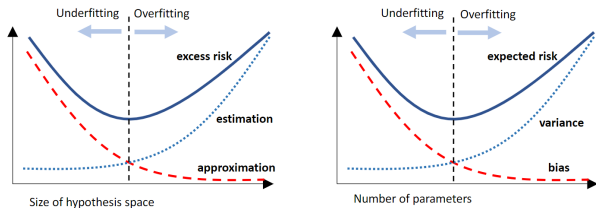
Learning goals

- Decomposing excess risk
- Into estimation, approx. and optim. error

- BV decomp often confused with related (but different) decomp:

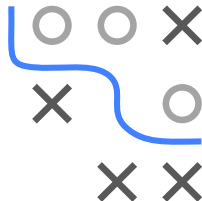
$$\underbrace{\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f_{\mathcal{H}_{all}}^*)}_{\text{excess risk}} = \underbrace{\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f_{\mathcal{H}}^*)}_{\text{estimation error}} + \underbrace{\mathcal{R}(f_{\mathcal{H}}^*) - \mathcal{R}(f_{\mathcal{H}_{all}}^*)}_{\text{approx. error}}$$

- Both commonly described using same figure and analogies



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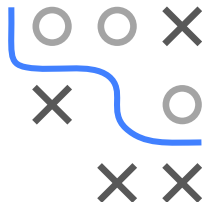
- BV decomp. only holds for certain losses, above is universal



- Approx. error is a structural property of $\mathcal{H}$
- Estimation error is random due to dependence on data in $\hat{f}$
- Estimation error occurs as we choose $f \in \mathcal{H}$ with limited train data minimizing $\mathcal{R}_{\text{emp}}$ instead of $\mathcal{R}$
- Knowing $\hat{f}_{\mathcal{H}} \in \arg \inf_{f \in \mathcal{H}} \mathcal{R}_{\text{emp}}(f)$ assumes we found a global minimizer of $\mathcal{R}_{\text{emp}}$, which is often impossible (e.g. ANNs)
- In practice, optimizing $\mathcal{R}_{\text{emp}}$ gives us “best guess” $\tilde{f}_{\mathcal{H}} \in \mathcal{H}$ of $\hat{f}_{\mathcal{H}}$
- Can now decompose its excess risk finer as

$$\underbrace{\mathcal{R}(\tilde{f}_{\mathcal{H}}) - \mathcal{R}(f_{\mathcal{H}_{\text{all}}}^*)}_{\text{excess risk}} = \underbrace{\mathcal{R}(\tilde{f}_{\mathcal{H}}) - \mathcal{R}(\hat{f}_{\mathcal{H}})}_{\text{optim. error}} + \underbrace{\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f_{\mathcal{H}}^*)}_{\text{estimation error}} + \underbrace{\mathcal{R}(f_{\mathcal{H}}^*) - \mathcal{R}(f_{\mathcal{H}_{\text{all}}}^*)}_{\text{approx. error}}$$

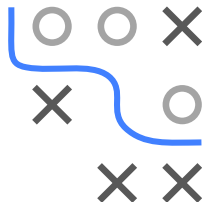
- NB: Optim err. can be < 0 , but $\mathcal{R}_{\text{emp}}(\tilde{f}_{\mathcal{H}}) \geq \mathcal{R}_{\text{emp}}(\hat{f}_{\mathcal{H}})$ always



- We can further decompose estimation error more finely by defining the *centroid* model or “systematic” model part
- For $\hat{f}_{\mathcal{H}} \in \arg \min_{f \in \mathcal{H}} \mathcal{R}_{\text{emp}}(f)$ centroid model under L2 loss is mean prediction at each x over all $\mathcal{D}_n$, $f_{\mathcal{H}}^{\circ} := \mathbb{E}_{\mathcal{D}_n \sim \mathbb{P}_{xy}^n} [\hat{f}_{\mathcal{H}}]$
- With $f_{\mathcal{H}}^{\circ}$, can decompose expected estimation error as

$$\underbrace{\mathbb{E}_{\mathcal{D}_n \sim \mathbb{P}_{xy}^n} [\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f_{\mathcal{H}}^*)]}_{\text{expected estimation error}} = \underbrace{\mathbb{E}_{\mathcal{D}_n \sim \mathbb{P}_{xy}^n} [\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f_{\mathcal{H}}^{\circ})]}_{\text{estimation variance}} + \underbrace{\mathcal{R}(f_{\mathcal{H}}^{\circ}) - \mathcal{R}(f_{\mathcal{H}}^*)}_{\text{estimation bias}}$$

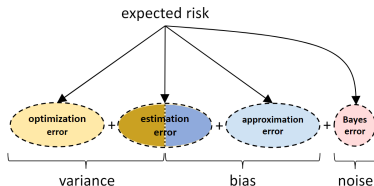
- Estimation bias measures distance of centroid model to risk minimizer over $\mathcal{H}$
- Estimation var. spread of ERM around centroid model induced by randomness due to $\mathcal{D}_n$



- Can now connect derived quantities back to bias and variance
- Bias is not only approx. error and variance is not estimation error
- Many details skipped here, see paper!

bias = approximation error + estimation bias

variance = optimization error + estimation variance



► Click for source

- **NB:** For special case of LM and L2 loss, we have very small optim / numerical error and estimation bias; so both decomp agree

